Enterprise Architect Thinking Guide — Cloud Migration Strategy

Author: Pushparaj Naik

Purpose: Interview preparation — Enterprise Architect mindset for cloud strategy, migration, and execution

Key Principle: *An Enterprise Architect doesn't think in tools — they think in business outcomes, risk, and trade-offs.*

Question 1: How Do You Decide Which Cloud Is Best Suited for Deployment/Migration?

The EA Mindset

An Enterprise Architect never says "I prefer AWS" or "We should use GCP." Instead, they say:

> *"The right cloud depends on the workload profile, organizational capabilities, regulatory constraints, and total cost of ownership. Let me walk you through how I evaluate this systematically."*

Framework: Cloud Selection Decision Matrix

I use a **weighted scoring model** across 12 dimensions, scored by a cross-functional team (Architecture, Security, Finance, Operations, Development):

#	Dimension	Weight	AWS	GCP	Azure	Scoring Criteria
1	**Workload Fit**	15%	9	8	9	Does the cloud have native services for our workload type?
2	**Existing Skills**	12%	Score based on team assessment			How many engineers are certified/experienced?
3	**Enterprise Readiness**	12%	9	7	9	Multi-account governance, IAM maturity, enterprise support
4	**Security & Compliance**	12%	9	8	9	Compliance certifications, security tooling depth
5	**TCO (3-year)**	10%	Based on pricing calculator			Compute + storage + network + support + licensing
6	**Data Residency**	10%	8	7	8	Region availability in required geographies
7	**Ecosystem & Marketplace**	8%	9	7	8	ISV integrations, marketplace maturity

#	Dimension	Weight	AWS	GCP	Azure	Scoring Criteria
8	**Hybrid/Multi-cloud**	6%	7	8	9	On-prem connectivity (Outposts, Anthos, Arc)
9	**Data & Analytics**	5%	8	9	8	Data platform services (Redshift vs BigQuery vs Synapse)
10	**AI/ML Capabilities**	4%	8	9	8	ML platform maturity (SageMaker vs Vertex AI vs Azure ML)
11	**Container/K8s**	3%	9	9	8	Managed Kubernetes maturity
12	**Vendor Relationship**	3%	Score based on existing contracts			Existing EDP, credits, account team engagement

Scoring: 1-10 per dimension. Final score = Σ (Weight × Score)

When Each Cloud Wins — Industry Patterns

Based on Gartner Magic Quadrant 2024-2025, Flexera State of the Cloud Report 2025, and Forrester Wave analysis:

Scenario	Best Fit	Why
Enterprise with large Windows/.NET estate	Azure	Native AD integration, SQL Server licensing benefits, Hybrid via Arc
Startup / Cloud-native, container-first	AWS or GCP	EKS/GKE maturity, serverless depth, developer experience
Heavy data analytics / ML-first	GCP	BigQuery is best-in-class for analytics, Vertex AI for ML pipelines
Banking / Financial Services	AWS	Deepest compliance portfolio (PCI-DSS, SOX), most regions, largest enterprise

Scenario	Best Fit	Why
		customer base
Government / Public Sector (India)	AWS	GovCloud equivalent, RBI compliance, 2 India regions (Mumbai, Hyderabad)
Multi-cloud strategy (mandated)	GCP (Anthos) or Azure (Arc)	Best multi-cloud management planes
SAP migration	AWS or Azure	Both have certified SAP partnerships; Azure has tighter SAP integration
IoT / Edge	AWS	IoT Core + Greengrass — most mature IoT platform
Media / Content delivery	AWS	CloudFront + MediaLive + S3 — largest CDN

Supporting Research & Artifacts

1. Gartner Magic Quadrant for Cloud Infrastructure & Platform Services (2024)

- AWS: Leader (furthest right for completeness of vision)
- Azure: Leader (close second, strongest hybrid story)
- GCP: Leader (strongest in data/AI, gaining enterprise traction)

2. Flexera 2025 State of the Cloud Report (key data points):

- 89% of enterprises have a multi-cloud strategy
- AWS leads in enterprise adoption (73%), Azure close (72%), GCP growing (38%)
- Top cloud challenge: managing cloud spend (82% of respondents)
- Top cloud initiative: migrating more workloads (65%)

3. Forrester Total Economic Impact Studies:

- AWS Migration: 3-year ROI of 241%, payback in < 6 months (Forrester TEI 2023)
- Average infrastructure cost reduction: 31% after migration
- Average downtime reduction: 69% after migration

4. IDC Cloud Pulse Survey 2024:

- 76% of enterprises cite security as top cloud selection criterion

- 68% cite total cost of ownership
- 54% cite existing team skills

Decision Tree (How I Present to Stakeholders)

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START: What's the primary workload?
│
├── Windows/.NET/SQL Server heavy?
│   └── Azure (licensing benefits via AHUB, native AD integration)
│
├── Data Analytics / ML first?
│   └── GCP (BigQuery + Vertex AI — best price-performance for analytics)
│
├── General enterprise / Microservices / IoT?
│   └── AWS (broadest service portfolio, deepest enterprise governance)
│
├── Regulatory mandate (data residency)?
│   └── Which cloud has regions in required geography?
│       ├── India only → AWS (2 regions) or Azure (3 regions)
│       └── Global → All three qualify
│
└── Multi-cloud mandated?
    └── Primary + secondary strategy
        └── Pick primary based on workload fit, secondary for specific use
cases
```

How I'd Answer in an Interview

> *"I don't pick a cloud based on personal preference. I run a structured evaluation with a weighted scoring model involving Architecture, Security, Finance, and Ops stakeholders. The evaluation covers 12 dimensions — workload fit, existing skills, compliance requirements, TCO, and ecosystem maturity.*
>
> *In my experience, AWS wins for general enterprise workloads and regulated industries because of its service breadth, compliance depth, and enterprise governance tooling. But if a client has a heavy .NET estate, Azure's licensing advantages make it the clear choice. And for data-first organizations, GCP's BigQuery is genuinely best-in-class.*
>
> *The key EA principle: the cloud is a business decision, not a technology decision. I present trade-offs with data, and let the business decide."*

Question 2: How Do You Convince Your CTO? Assessment Strategy for Migrating 1000 Servers

The EA Mindset

The CTO cares about three things:

1. **Risk:** "Will this break anything?"
2. **Cost:** "How much will it cost, and when do we see ROI?"
3. **Timeline:** "How long will this take?"

Your job as EA is to answer all three with **data, not opinions.**

Phase 1: Discovery & Assessment (Weeks 1-6)

Tooling:

- **AWS Migration Hub + Application Discovery Service** — agent-based discovery of all 1000 servers
- **TSO Logic / Migration Evaluator** (free AWS tool) — generates TCO comparison

What We Discover:

Data Point	Why It Matters
Server specs (CPU, RAM, disk, OS)	Right-sizing recommendations
Network dependencies (server-to-server communication)	Migration wave grouping
Application mapping (which servers form an application)	Prevents breaking app dependencies
Utilization data (30-day average CPU/memory)	Right-sizing — most servers are 20-30% utilized

Data Point	Why It Matters
License inventory (Oracle, SQL Server, SAP, etc.)	Licensing strategy — BYOL vs included vs replace
Data volumes and growth rate	Storage and transfer cost estimation

Output: Migration Portfolio Analysis

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1000 Servers Assessed:
├── 300 servers (30%) — Retire/Decommission
│   └── Unused, redundant, or end-of-life — immediate savings
├── 400 servers (40%) — Rehost (Lift & Shift)
│   └── Move to EC2 with minimal changes — fastest migration
├── 200 servers (20%) — Replatform
│   └── Move to managed services (RDS, ECS, ElastiCache)
├── 80 servers (8%) — Refactor
│   └── Modernize to containers/serverless — highest long-term value
└── 20 servers (2%) — Retain
    └── Regulatory or technical constraints — stay on-prem
```

Phase 2: CTO Presentation — The Business Case

Slide 1: Current State Pain Points

Pain Point	Cost/Impact
Data center lease renewal (18 months out)	$2.4M/year
Hardware refresh cycle needed	$3.5M capital expenditure
Average provisioning time	6-8 weeks (vs. minutes in cloud)
Last year's downtime	47 hours (costing ~$500K in lost productivity)
DR capability	Backup only, no tested DR — RTO = 72+ hours

Slide 2: TCO Comparison (3-Year)

Item	On-Premises (3yr)	AWS (3yr)	Savings
Compute (700 servers after retire)	$4.2M	$2.1M (right-sized + Savings Plans)	50%
Storage	$1.8M	$0.6M (S3 tiering + EBS optimization)	67%
Networking	$0.9M	$0.4M	56%
Licensing (SQL Server, Windows)	$2.1M	$1.4M (AHUB-equivalent, Aurora migration)	33%
Data center / Facilities	$7.2M	$0	100%
Staff (reduced ops overhead)	$3.6M	$2.4M (automation reduces headcount needs)	33%
Migration cost (one-time)	$0	**$1.8M**	—
Total	**$19.8M**	**$8.7M**	**56% savings**
3-Year Net Savings	—	—	**$11.1M**

Slide 3: Risk-Adjusted Timeline

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Month 1-2:  Foundation (Landing Zone, networking, security baseline)
Month 3-4:  Wave 1 — Non-critical apps (50 servers) — build confidence
Month 5-8:  Wave 2 — Business apps (200 servers) — validated patterns
Month 9-14: Wave 3 — Core systems (350 servers) — highest risk, most prep
Month 15-16: Wave 4 — Database migrations (100 servers) — DMS + cutover
Month 16-18: Decommission on-prem, optimize cloud
```

Total: 18 months for 1000 servers (including 300 retirements)

Migration Wave Strategy (The Key to Cost Control)

Wave Grouping Criteria:

Factor	How It Affects Waves
Application dependencies	Servers that talk to each other move together
Business criticality	Low-risk first, critical last (build confidence)
Complexity	Simple rehost first, complex refactor last
Business calendar	Avoid moves during peak business periods
Team capacity	50-80 servers per wave (manageable batch size)

Cost Control Levers:

1. **Retire 30% upfront** — saves $0 migration cost for 300 servers
2. **Right-size everything** — most servers are over-provisioned; average 40% cost reduction
3. **Reserved Instances / Savings Plans** — commit after Wave 1 for proven workloads (30-40% savings)
4. **Managed services** — replace 200 self-managed databases with RDS (reduce DBA effort by 60%)
5. **Automation** — invest in Terraform + CI/CD upfront; pay off by Wave 2
6. **AWS Migration Credits** — apply for MAP program (up to $100K-500K in credits for large migrations)

How to Convince the CTO

Don't lead with technology. Lead with business risk:

> *"We have an 18-month window before our data center lease renewal. If we don't act, we're committing to another 5-year, $12M lease plus a $3.5M hardware refresh. Cloud migration costs $1.8M one-time and saves us $11M over 3 years. The risk of NOT migrating is greater than the risk of migrating."*

Then address their fears:

CTO Fear	Your Response
"What if it costs more than expected?"	"We'll run a 50-server pilot wave first. If unit costs are > 10% above estimate, we pause and re-baseline."

CTO Fear	Your Response
"What about downtime during migration?"	"We use AWS DMS with CDC for databases — zero downtime cutover. Applications use blue-green DNS switch."
"Do we have the skills?"	"We'll embed an AWS Professional Services architect for Wave 1. By Wave 2, our team is self-sufficient."
"What if cloud is MORE expensive long-term?"	"Our TCO includes Savings Plans, right-sizing, and managed services. I'll set up monthly FinOps reviews to track actuals vs. forecast."